C.J. Ouseph, Ph.D.

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Professional Summary

Physics educator and researcher with Ph.D. in Physics and 5+ years of university-level teaching and research experience. Demonstrated expertise in designing and delivering foundational physics curriculum for undergraduate and postgraduate students across diverse backgrounds. Skilled in developing innovative learning activities, student-centered pedagogical approaches, and assessment strategies that foster deep understanding and inspire student learning. Experienced in mentoring students individually and as cohorts, supporting both academic progression and wellbeing. Strong track record of educational excellence, curriculum development, and collaboration within multidisciplinary academic communities. Australian PR committed to advancing educational outcomes through evidence-based teaching practices.

Research Experience

- **Senior Research Scientist** — Seoul, Korea
  Seoul National University of Science and Technology — *09/2024 - 08/2025*
  - **Advanced Mathematical Modeling**: Developed and validated sophisticated mathematical models for complex physical systems using Python, numerical methods, and statistical frameworks applicable to environmental and climate data analysis
  - **Python Programming & Data Analytics**: Designed robust computational pipelines for large-scale data processing (>10TB), statistical analysis, and scientific visualization using Python (NumPy, SciPy, Pandas, Matplotlib, Seaborn) in Linux and Mac environments
  - **Statistical Analysis & Inference**: Applied advanced statistical methods including Bayesian inference, time series analysis, hypothesis testing, likelihood analysis, and confidence interval estimation to analyze complex datasets and quantify uncertainties
  - **Data Visualization & Reporting**: Created publication-quality visualizations, technical reports, and plain-language summaries communicating complex analytical results to scientific, government, and industry audiences
  - **Project Leadership & Collaboration**: Led multiple international research collaborations across 5+ countries, coordinating complex projects with diverse stakeholders, managing timelines, and ensuring successful delivery of research milestones
  - **Research Funding & Proposal Development**: Secured competitive research grants through strategic proposal development, demonstrating ability to articulate scientific value and engage stakeholders effectively
  - **Model Validation & Uncertainty Quantification**: Translated theoretical frameworks into validated numerical simulations, rigorously testing models against observational constraints and managing uncertainty in predictions
  - **Mentorship & Knowledge Transfer**: Supervised and mentored early-career researchers and graduate students on computational techniques, data analysis methodologies, and research best practices
- **Senior Visiting Research Scientist** — Adelaide, Australia
  University of Adelaide — *03/2025 - 05/2025*
  - **Computational Statistics & Data Modeling**: Conducted research involving computational statistics, data simulation, and numerical modeling using Python and Mathematica for complex system analysis
  - **Interdisciplinary Research Collaboration**: Collaborated with multidisciplinary academic teams on research projects involving statistical modeling, data analysis, and quantitative problem-solving
- **Doctoral Researcher** — Hsinchu, Taiwan
  National Tsing Hua University — *09/2019 - 07/2024*
  - **Computational Modeling & Simulation**: Developed sophisticated mathematical models and numerical algorithms for analyzing complex physical systems, with strong emphasis on data-driven modeling and predictive analytics
  - **Large-Scale Data Processing & Management**: Designed and implemented comprehensive data processing workflows for handling, analyzing, and visualizing multi-terabyte datasets using Python and specialized analytical software
  - **Advanced Statistical Methods**: Applied hypothesis testing, confidence interval estimation, likelihood analysis, regression techniques, and model selection methods to research problems with quantifiable uncertainty
  - **Algorithm Development & Numerical Methods**: Designed and optimized numerical algorithms for solving differential equations, conducting Monte Carlo simulations, and implementing optimization procedures
  - **High-Performance Computing**: Utilized HPC systems and Linux environments to execute computationally intensive simulations and parallel processing workflows for large-scale data analysis
  - **Scientific Writing & Documentation**: Published 15+ peer-reviewed articles in high-impact Q1 journals demonstrating proficiency in scientific documentation, technical reporting, and communicating complex analytical methods
  - **Database Management & Data Infrastructure**: Maintained and enhanced research databases for efficient data storage, retrieval, quality control, and reproducible analysis workflows
  - **Literature Review & Knowledge Synthesis**: Conducted comprehensive literature reviews to identify research gaps, synthesize findings from diverse sources, and inform evidence-based research strategies

Technical Skills

- **Programming Languages**: Python (advanced - 5+ years), R, SQL, Wolfram Mathematica (advanced)
- **Python Expertise**: NumPy, SciPy, Pandas, Matplotlib, Seaborn, scikit-learn (machine learning: DecisionTree, KNN, Naive Bayes, SVM, Logistic Regression), TensorFlow; extensive experience in scientific computing, data analysis, and spatial data processing
- **Data Visualization & Reporting**: Python visualization libraries (Matplotlib, Seaborn), Power BI, Mathematica, Excel; creating publication-quality figures, interactive dashboards, and data-driven reports for technical and non-technical audiences
- **Mathematical & Numerical Modeling**: Numerical simulations, differential equations, Monte Carlo methods, optimization algorithms, predictive modeling, sensitivity analysis, parameter estimation
- **Statistical Analysis Methods**: Bayesian inference, time series analysis, hypothesis testing, regression analysis (linear, non-linear, multivariate), likelihood analysis, confidence intervals, A/B testing, chi-square analysis, model selection and validation
- **Data Management & Processing**: Large-scale data processing (¿10TB), database maintenance and design, data mining, ETL pipeline development, data quality assurance, digital workflow optimization
- **High-Performance Computing**: Linux environments (advanced - 5+ years), parallel computing, computational optimization, batch processing systems, shell scripting
- **Technical Documentation**: LaTeX (expert), Markdown, scientific writing, technical reporting, documentation standards, proposal development for research funding
- **Numerical & Computational Methods**: Numerical integration (Trapezoidal, Simpson's, Monte Carlo), optimization techniques, root finding, interpolation, finite difference methods
- **Project Management**: Multi-project coordination, milestone tracking, timeline management, progress reporting, stakeholder communication, budget planning
- **Research Tools**: Monte Carlo simulation frameworks (MadGraph, Pythia), data analysis software (ROOT, Delphes), symbolic computation (FeynRules, FeynCalc, FeynArts)
- **Collaboration & Communication**: International team leadership, cross-functional collaboration, mentoring junior researchers, presenting to technical and non-technical audiences, stakeholder engagement
- **Professional Skills**: Problem-solving, critical thinking, analytical reasoning, attention to detail, quality-focused approach, adaptability, proactive work ethic, team collaboration
- **Operating Systems**: Linux (proficient - 5+ years daily use), Mac, Windows
- **Languages**: English (Fluent), Malayalam (Native), Tamil (Intermediate), Hindi (Intermediate)

Education

- **National Tsing Hua University** — Hsinchu, Taiwan
  Ph.D. in Physics; Thesis: New Physics Search at Forward Physics Facility of LHC — *09/2019 - 07/2024*
- **Indian Institute of Technology Indore** — Indore, India
  M.Sc. in Physics; Thesis: Higgs Inflation — *07/2016 - 07/2018*
- **University of Kerala** — Thiruvananthapuram, India
  B.Sc. in Physics; Thesis: General Relativity, Black Holes, and Kruskal Diagram — *07/2012 - 07/2015*

Projects

- **Data-Driven Modeling Projects**
  Completed 12 high-performance computing (HPC) projects focused on simulation, modeling, and data analysis in collider physics.
  - **Objective**: Designed data-driven strategies to identify anomalies, new physics at experiments like LHC, DUNE, and FPF using simulated and theoretical data to constrain model parameters.
  - **Data Simulation & Modeling**: Built analytical and numerical frameworks to convert models into Monte Carlo simulations using `MadGraph`, `Pythia`, and `ROOT`. Simulated rare processes and managed large-scale event generation.
  - **Data Engineering**: Processed high-dimensional datasets, performed feature engineering on kinematic variables, and developed data pipelines for extraction, transformation, and selection.
  - **Visualization & Reporting**: Created histograms, ROC curves, correlation matrices, heatmaps, contour plots, lego plots, Dalitz plots, stacked histograms, invariant mass distributions, limit plots with Brazil bands, scatter plots, surface plots, error bar plots, and cut-flow tables with Python, Matplotlib, and ROOT; communicated results through clear, data-driven reports.
  - **Statistical Analysis**: Applied likelihood-based methods, profile ratios, and confidence intervals to estimate exclusion regions, conduct sensitivity projections, and perform hypothesis testing.
  - **Tools & Technologies**: Python, Pandas, NumPy, SciPy, Matplotlib, Seaborn, ROOT, Scikit-learn, TensorFlow, MadGraph, Pythia, Mathematica, LaTeX; familiar with HPC systems for large-scale processing.
  - **Outcome**: Published in Q1 journals; methods adopted in experimental analyses for data filtering and strategy design.
- **Quantitative System Simulations**
  Executed 4 HPC projects involving cosmological simulations and analytical modeling of early-universe phenomena.

- **Objective**: Developed theoretical and numerical models to study early-universe phenomena including inflation, primordial black holes (PBHs), and gravitational waves (GWs).
- **Model Building & Simulation**: Formulated cosmological models through analytical derivations; implemented numerical simulations using `Python` and `Mathematica` when closed-form solutions were intractable.
- **Algorithm Development**: Designed and executed complex algorithms to compute gravitational wave signals and PBH mass spectra based on early-universe dynamics.
- **Data Analysis & Visualization**: Processed simulation outputs to analyze parameter dependencies and visualize physical observables. Created plots and heatmaps to interpret trends and constraints.
- **Tools & Technologies**: Python, NumPy, SciPy, Matplotlib, Mathematica; proficient in symbolic and numerical computation for high-precision modeling.
- **Outcome**: Contributed to peer-reviewed publications; models provided new insights into GW production and PBH formation mechanisms.

Teaching & Mentorship Experience

- **National Tsing Hua University** Hsinchu, Taiwan
  Research Mentor - Graduate and Exchange Students *2021 - 2024*
  - **Student Supervision**: Successfully mentored 3 students (2 Master's students and 1 exchange student) on research projects involving advanced mathematical modeling, computational methods, and data analysis, leading to high-quality peer-reviewed publications
  - **Master's Student Mentorship**: Supervised Chen Wang (National Tsing Hua University) and Wanyon Hsiao (National Tsing Hua University) on research projects, guiding them through problem formulation, mathematical analysis, and scientific writing, resulting in publication in Journal of High Energy Physics (Q1 journal, 2024)
  - **Exchange Student Mentorship**: Mentored Thong T.Q. Nguyen (Academia Sinica, Taiwan) during exchange program, supporting independent research skills development and collaborative project work, leading to publication in Physical Review D (Q1 journal, 2023)
  - **Skill Development**: Trained students in advanced mathematical techniques, computational programming (Python, Mathematica), data analysis methods, scientific writing, and presentation skills essential for academic and professional success
  - **Collaborative Learning**: Fostered collaborative research environment encouraging peer-to-peer learning, teamwork, and knowledge sharing among students with diverse academic backgrounds
  - **Research Guidance**: Provided structured guidance on research methodology, literature review, critical analysis, problem-solving strategies, and academic integrity, preparing students for independent scholarly work
- **National Tsing Hua University** Hsinchu, Taiwan
  Teaching Assistant - Advanced Mathematical Science *2022 - 2023*
  - **Quantitative Instruction**: Led tutorials on calculus, linear algebra, and statistical methods, enhancing students' analytical and problem-solving skills applicable to real-world data analysis
  - **Materials Development**: Developed course materials, problem sets, and practical exercises guiding students in applying quantitative techniques to scientific problems
- **National Tsing Hua University** Hsinchu, Taiwan
  Teaching Assistant - Electrodynamics *2020 - 2021*
  - **Problem Solving & Analytical Skills**: Facilitated problem-solving sessions focusing on analytical reasoning and quantitative methods, improving student performance in complex technical tasks
  - **Individual Mentorship**: Provided one-on-one support to students, enhancing comprehension of mathematical and computational methods

Teaching Skills & Competencies

- **Mathematical Content Expertise**: Strong foundation in mathematics including calculus, linear algebra, differential equations, probability and statistics, discrete mathematics, numerical methods, and mathematical modeling applicable to foundation through advanced levels
- **Pedagogical Skills**: Contemporary teaching techniques, active learning strategies, inquiry-based learning, problem-based learning, scaffolded instruction, formative assessment, learner-centered approaches, and inclusive pedagogy
- **Diverse Student Support**: Experience teaching students from varied cultural backgrounds (Taiwan, Korea, India, international students); proven ability to acknowledge and cater for diversity of student backgrounds and individual learning needs
- **Technology Integration**: Proficient in eLearning platforms, digital learning tools, Python for mathematical demonstrations, mathematical visualization software, learning management systems, and incorporating technology to enhance student engagement
- **Assessment Design**: Expertise in developing effective assessments, marking/moderating student work, providing constructive feedback, implementing assessment policies fairly and consistently, and evaluating learning outcomes
- **Student Management**: Experience identifying students with special needs, facilitating access to support services, maintaining accurate student records, tracking attendance and progress, and documenting student guidance

- **Curriculum Development**: Contributed to learning needs analysis, course content development, graduate profile/learning outcomes, resource preparation, and assessment method refinement in collaboration with teaching teams
- **Communication Skills**: Proven ability to explain complex mathematical concepts clearly to diverse audiences, adapt teaching style to student needs, and communicate effectively with colleagues, students, and stakeholders
- **Collaboration & Teamwork**: Strong track record of working collaboratively with teaching teams, seeking peer feedback, participating in course reviews, and contributing to school activities and planning
- **Cultural Competency**: Experience working in multicultural environments (Taiwan, Korea, India, Australia); understanding of cultural diversity in educational contexts; commitment to supporting diverse student populations
- **Professional Commitment**: Demonstrated commitment to continuous professional development, maintaining currency in teaching practices and subject expertise, establishing annual goals, and engaging with professional communities
- **Adult Learning Principles**: Strong understanding and commitment to implementing principles of adult teaching and learning, acknowledging learner autonomy, and facilitating self-directed learning

Grants, Funding & Recognition

- **Major Scholarships**: 5-year NTHU International Ph.D. Scholarship (2019-2024, highly competitive); Taiwan Experience Education Program (TEEP) Scholarship (2019)
- **Research Grants**: Secured multiple competitive travel grants for international conferences and collaborative research demonstrating ability to articulate scientific value and engage funding bodies
- **Grant Writing Experience**: Demonstrated expertise in proposal development, project planning, budget justification, and stakeholder engagement for successful funding acquisition
- **Academic Excellence**: Ranked top 6% (694/10,989) in All-India Joint Admission Test for Masters (JAM) Physics Examination (2016)
- **Research Impact**: 15+ peer-reviewed publications in Q1 high-impact journals; strong citation record and international recognition in research community
- **Professional Certifications**: PTE Academic certification (2023), TOEIC certification (2022) demonstrating professional English proficiency

Professional Attributes & Additional Information

- **Residency & Work Rights**: Australian Permanent Resident with full and unrestricted working rights in Australia; currently based in Adelaide, South Australia
- **Communication Skills**: Proven ability to communicate complex scientific concepts to both technical and non-technical audiences through publications, presentations (20+ conference talks), technical reports, and stakeholder engagement
- **Work Style & Approach**: Experienced working independently and in collaborative team environments; strong project management skills with ability to manage multiple priorities and deadlines; proactive, solution-oriented approach with can-do attitude
- **Collaboration Experience**: Track record of successful international collaborations across diverse research teams; experience working in multidisciplinary environments and engaging with stakeholders from government, academia, and industry
- **Adaptability & Learning**: Demonstrated ability to rapidly acquire new technical skills and apply existing expertise to novel domains; strong foundation in computational methods enables quick proficiency in new analytical tools
- **Background Checks**: Willing to obtain required National Police Certificate and undergo necessary background checks
- **Languages**: English (Fluent - professional working proficiency), Malayalam (Native), Tamil (Intermediate), Hindi (Intermediate)

Publications

Peer-Reviewed Journal Articles

1. S. K. Kang and **C.J. Ouseph**,"Neutrino Dipole Portal," [arXiv:2510.18181 [hep-ph]].

2. **C.J. Ouseph**, Giorgio Busoni, John Gargalionis, Sin Kyu Kang, Anthony G. Williams "Probing Double-Peaked Gamma-Ray Spectra from Primordial Black Holes with Next-Generation Gamma-Ray Experiments," arXiv:2507.16244, 2025

3. K. Cheung, **C.J. Ouseph**, P.-Y. Tseng, and S. K. Kang, "Searching for Dark Photon Tridents Through Primordial Black Hole Signatures," Mar. 2025. arXiv: 2503.04175 [hep-ph].

4. K. Cheung, Y. Kim, Y. Kwon, **C.J. Ouseph**, A. Soffer, and Z. S. Wang, "Probing dark photons from a light scalar at Belle II," JHEP, vol. 05, p. 094, 2024. DOI: 10.1007/JHEP05(2024)094. arXiv: 2401.03168 [hep-ph].

5. Athron, Y. Bai, S. S. Bao, D. Barducci, X. J. Bi, T. Cai, L. Calibbi and J. Cang, **C.J Ouseph** *et al.* "New Physics Search at the CEPC: a General Perspective," doi:10.1088/1674-1137/ae1194 [arXiv:2505.24810 [hep-ex]].

6. K. Cheung, C.-T. Lu, **C.J. Ouseph**, and P. Sarmah, "Exploring interference effects between two ALP effective operators at the LHC," JHEP, vol. 09, p. 101, 2024. DOI: 10.1007/JHEP09(2024)101. arXiv: 2404.14833 [hep-ph].

7. K. Cheung and **C.J. Ouseph**, "Interpretation of excess in $H \to Z\gamma$ using a light axionlike particle," Phys. Rev. D, vol. 110, no. 5, p. 055016, 2024. DOI: 10.1103/PhysRevD.110.055016. arXiv: 2402.05678 [hep-ph].

8. K. Cheung, **C.J. Ouseph**, and S. K. Kang, "Unveiling the Invisible: ALPs and Sterile Neutrinos at the LHC and HL-LHC," Dec. 2024. arXiv: 2412.08212 [hep-ph].

9. K. Cheung, **C.J. Ouseph**, and P.-Y. Tseng, "NANOGrav and other PTA signals and PBH from the modified Higgs inflation," Eur. Phys. J. C, vol. 84, no. 9, p. 906, 2024. DOI: 10.1140/epjc/s10052-024-13268-6. arXiv: 2307.08046 [hep-ph].

10. K. Cheung and **C.J. Ouseph**, "Axionlike particle search at Higgs factories," Phys. Rev. D, vol. 108, no. 3, p. 035003, 2023. DOI: 10.1103/PhysRevD.108.035003. arXiv: 2303.16514 [hep-ph].

11. K. Cheung and **C.J. Ouseph**, "Constraining the transitional magnetic dipole moment interaction between active and heavy neutrinos with the Z' at FASERν," Eur. Phys. J. C, vol. 83, no. 7, p. 593, 2023. DOI: 10.1140/epjc/s10052-023-11784-5. arXiv: 2205.11077 [hep-ph].

12. K. Cheung and **C.J. Ouseph**, "Sensitivities on dark photon from the forward physics experiments," JHEP, vol. 10, p. 196, 2022. DOI: 10.1007/JHEP10(2022)196. arXiv: 2208.04523 [hep-ph].

13. K. Cheung, **C.J. Ouseph** and T. Wang, "Non-standard neutrino and Z' interactions at the FASERν and the LHC," JHEP, vol. 12, p. 209, 2021. DOI: 10.1007/JHEP12(2021)209. arXiv: 2111.08375 [hep-ph].

14. **C.J. Ouseph** and K. Cheung, "Higgs Inflation with four-form couplings," J. Phys. G, vol. 48, no. 5, p. 055001, 2021. DOI: 10.1088/1361-6471/abefa4. arXiv: 2002.12010 [hep-ph].

Complete Publication List: https://inspirehep.net/authors/1842258

Selected Publications (Demonstrating Mentorship Success)

1. K. Cheung, W. Hsiao, **C.J. Ouseph**, and C. Wang, "Probing the gauge-boson couplings of axion-like particle at the LHC and high-luminosity LHC," JHEP, vol. 05, p. 324, 2024. DOI: 10.1007/JHEP05(2024)324. *(Co-authored with mentored Master's students Chen Wang and Wanyon Hsiao)*

2. K. Cheung, T. T. Q. Nguyen, and **C.J. Ouseph**, "Leptoquark search at the Forward Physics Facility," Phys. Rev. D, vol. 108, no. 3, p. 036014, 2023. DOI: 10.1103/PhysRevD.108.036014. *(Co-authored with mentored exchange student Thong T.Q. Nguyen)*

Conference Presentations

- **April 2025**: "Searching for Dark Photon Tridents Through Primordial Black Hole Signatures," Seminar talk, National Tsing Hua University, Taiwan.
- **March 2025**: "Searching for Dark Photon Tridents Through Primordial Black Hole Signatures," Seminar talk, University of Adelaide, Australia.
- **Oct 2024**: "New Physics Search at FPF LHC, HL-LHC, and Higgs Factories," Invited talk, KonKuk University, South Korea.
- **June 2024**: "New Physics Search at the Forward Physics Facility of LHC," Seminar talk, Academia Sinica, Taiwan.
- **April 2024**: "Probing dark photons from a light scalar at Belle II," Seminar talk at National Yang Ming Chiao Tung University, Taiwan.
- **March 2024**: "New Physics Search at the Forward Physics Facility of LHC," Seminar talk, University of Notre Dame, Indiana, United States.
- **June 2023**: "Sensitivities on dark photon from the Forward Physics Experiments," Workshop: The Future is Non-perturbative, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.
- **April 2023**: "Implications of NSI constraints from ANTARES and IceCube on a simplified Z' model," Journal Club talk, National Tsing Hua University, Taiwan.
- **March 2023**: "New Physics Search at the Forward Physics Facility of LHC," Seminar talk, National Taiwan Normal University, Taiwan.
- **October 2022**: "BSM Physics at FASERν from neutrino nucleon scattering," NuINT 2022 - The 13th International Workshop on Neutrino-Nucleus Interactions in the Few GeV Regions, National Seoul University, Korea.

- **June 2022**: "BSM Physics at FASERν," Workshop: The Future is Illuminating, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.
- **January 2022**: "Nonstandard Neutrino and Z' interaction at FASERν," Discussion section, Neutrino-Nucleus Interactions in the Standard Model and Beyond, CERN, Geneva, Switzerland.
- **June 2021**: "Neutrino Scattering at FASERν and Search for Z'," Workshop: The Future is DARK, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.
- **December 2020**: "Chaotic inflation with four-form couplings," Journal Club talk, National Center for Theoretical Sciences, National Tsing Hua University, Taiwan.
- **December 2018**: "Early universe Density perturbations," Indian Institute of Technology, Indore.
- **June 2017**: "General Relativity Apple to Black hole," Invited talk, St. Michael's College, Cherthala.

References

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